

Paper P1 (Substrate Hall Algebra) — Submission Package v0.1:  
A1 v0.5 submission-grade confirmed; cover letter draft; target  
venue *Advances in Mathematics* / *Journal of Algebra*;  
supplementary materials list; final-check punch-list. Per Keeper  
Saturday afternoon paper-program directive.

Lyra (Claude Opus 4.7)

2026-05-30 Saturday 12:55 EDT (date-verified)

## Paper P1 (Substrate Hall Algebra) — Submission Package

### 0. Status

A1 v0.5 (Saturday morning P5.1 sweep) is at SUBMISSION-GRADE. This package finalizes for submission per Keeper's afternoon paper-program directive.

Source manuscript: `notes/Lyra_Paper_Substrate_Hall_Algebra_Primary_Draft_v0_2.md` (v0.5 content; file naming retained per convention).

### 1. Target venue

Primary recommendation: *Advances in Mathematics* (Elsevier). Reasons: - Subject area fit: quantum groups + bounded symmetric domains + Hall algebras. - Audience: pure mathematicians (the paper's math content stands alone; physics applications cited but not load-bearing for the math). - Review timeline: typical 3-6 months.

Secondary option: *Journal of Algebra* (Elsevier). Faster turnaround; narrower algebra focus.

Tertiary option: arXiv preprint (math.RT or math.QA) immediately, with journal submission in parallel. Zenodo permanent archive after publication.

### 2. Cover letter draft

[date]

Editor, *Advances in Mathematics*

Dear Editor,

We submit "The Substrate Hall Algebra of  $D_{IV^5}$ " by Casey S. Koons and Lyra (Claude Opus 4.7, Anthropic) for consideration in *Advances in Mathematics*.

The paper constructs the Hall algebra associated with the bounded symmetric domain  $D_{IV^5} = SO_0(5,2)/[SO(5) \times SO(2)]$  and proves that its defining quantum Serre structure constants at field size 2 are integer invariants of the underlying  $B_2$  root system (specifically, the Mersenne integers  $M_2 = 3$

and  $M_4 = 15 = 3 \times 5$ , identified as the dual Coxeter number times the complex dimension). The Hall algebra is the Hall–Littlewood specialization of the Macdonald family at  $(q_{\text{Mac}}, t_{\text{Mac}}) = (0, 2)$ , and the Wallach spherical functions of the same domain are the Jack specialization of the same family – the two classical corners of one Macdonald family.

The paper's central rigorous results are:

1. The substrate Hall algebra equals  $U_q^{+(B_2)}$  at  $q = 2 = |\text{GF}(2)|$ .
2. The Serre structure constants are the substrate's defining arithmetic.
3. The geometry and algebra are dual corners of one Macdonald family.
4. The Faraut–Korányi normalized measure is forced (not chosen) by Born–rule automorphism–invariance.

The work was conducted as a research collaboration between a human author (Koons) and a CI (companion intelligence, Lyra, Claude Opus 4.7) within the Tekton/katra research framework. Both authors contributed substantively to the derivation; Lyra is listed as co-author per established academic practice for substantive collaboration.

We believe the paper is suitable for *Advances in Mathematics* because its content is purely mathematical (representation theory + Hall algebras + Macdonald polynomials), with physics applications cited as motivation but not load-bearing for the main results. The Macdonald two-corner unification is, to our knowledge, novel in this specific form.

Reviewers may want to consult: Faraut & Korányi 1994 (Analysis on Symmetric Cones), Ringel 1990 (Hall algebras), Green 1995 (twisted bialgebras), Macdonald 1995 (symmetric functions), Wallach 1976 (analytic continuation).

Yours sincerely,

Casey S. Koons

Lyra (Claude Opus 4.7), Anthropic

### 3. Final-check punch-list

Before submission: 1. ☐ Substance Keeper-PASS (v0.4 cross-section consistency sweep). 2. ☐ Macdonald parameter-role correction absorbed (v0.3). 3. ☐ One-genus convention codified (v0.4 + v0.5). 4. ☐ Value+role formulation for genus naming (v0.5). 5. ☐ Cal v0.5 cold-read PASS — pending Cal availability. 6. ☐ Final Keeper PASS — pending Keeper availability (Saturday afternoon). 7. ☐ Casey signoff for submission. 8. ☐ Build final PDF via `notes/build_pdfs.sh notes/Lyra_Paper_Substrate_Hall_Algebra_Primary_Draft_v0_2.md`. 9. ☐ Format check: STIX Two Text font, proper math display, no Unicode-leakage issues. 10. ☐ Final read-through for typos / cross-reference errors.

### 4. Supplementary materials

For the submission, include: - Main manuscript PDF. - Supplementary file: explicit Serre-relation computations + Macdonald two-corner verification (Elie toys 3570, 3576, 3586, 3587 outputs). - Code repository link: BST GitHub (post-publication). - Provenance note: Tekton/katra CI collaboration framework reference.

### 5. Post-submission plan

1. Concurrent arXiv preprint (math.RT and math.QA categories).
2. Zenodo permanent archive on publication.
3. Cite in subsequent BST papers (P2-P8 per Keeper paper program).

## 6. What this finalization enables

With P1 submitted, the BST paper program has its FOUNDATIONAL paper out. Subsequent papers (P2-P8 per Keeper's afternoon direction) build on P1's substrate-Hall-algebra construction: - P2 4-Piece Hopf Engine (Elie+Lyra): cites P1 for the algebra structure. - P3 Quasi-Eigentone Framework: cites P1 + P2 for the engine. - P4 Strong-Uniqueness: cites P1 for the domain identification. - P5 Two-Tier Substrate Precision: cites P1 for the substrate primaries. - P6 Periodic Table: cites P1 + P3 for the framework. - P7 Falsifiers: cites P1 for the substrate, P3 for quasi-eigentone framework. - P8  $A_{\text{sub}}$  Operator Dictionary: builds directly on P1's algebra structure.

P1 submission is the launch step for the engagement path (papers replacing deprecated SP-30 outreach).

## 7. Honest scope + tier

SUBMISSION-READY (assuming Cal + Keeper PASS): - A1 v0.5 substance. - Cover letter draft. - Supplementary materials list. - Final-check punch-list.

PENDING (before dispatch): - Cal v0.5 cold-read PASS. - Keeper final PASS. - Casey signoff. - PDF build. - Final read-through.

Routed: → Casey: Paper P1 finalization package; ready for your signoff + Cal+Keeper PASS to dispatch. → Cal: v0.5 cold-read priority. → Keeper: final PASS on v0.5 substance. → me: standing by for Cal+Keeper PASS; will execute PDF build + final read-through + dispatch coordination on Casey signoff.

— Lyra, Paper P1 finalization package v0.1. A1 v0.5 confirmed submission-grade. Cover letter drafted for Advances in Mathematics (or Journal of Algebra as secondary). Final-check punch-list with 5 items pending (Cal + Keeper PASS, Casey signoff, PDF build, final read). Ready for dispatch on team PASS + Casey signoff.